Standing at the frontier of foundation models and learning theory, I believe robotics has reached a turning point. I want to build the next generation of general-purpose robotic systems that are both intelligent and trustworthy, bridging human intelligence and physical intelligence within a unified framework. It is with this broad goal that I am applying to pursue a PhD, and I ultimately aspire to become a professor. Having received the **Best Undergraduate Teaching Fellows Award**, I also hope to advance equitable access to higher education and help as many students as possible thrive in engineering and research. Specifically, my research focuses on two complementary directions:

(1) **Human intelligence for physical intelligence**: modeling human goals, preferences, and decision-making to make embodied agents more robust, adaptive, and aligned with human needs.

(2) **Physical intelligence around humans**: designing control and learning methods that enable physical agents to act safely in human-centered environments, interacting with people in the reality.

Human intelligence for physical intelligence

Chinese University of Hong Kong, Shenzhen *Can visual cues help us understand spoken instructions better?* My research journey in human intelligence began with an everyday observation: humans can effortlessly focus on a specific speaker in a noisy, multi-speaker environment, whereas current AI agents struggle to identify and follow the intended speaker. This ability, often referred to as selective auditory attention, is well documented behaviorally but its multimodal underpinnings especially the role of vision remain insufficiently understood. To address this gap, I helped design controlled experiments to test whether synchronized visual cues (such as the speaker's face and gaze) improve listeners' ability to focus on and comprehend a target speaker. Our results show that synchronized visual and auditory input significantly enhances people's understanding of the spoken content. Under the supervision of Prof. Haizhou Li, our work received the **Best Student Paper Award** at the **2024 International Conference on Social Robotics (ICSR)** [1]. My contributions included co-designing and implementing the experimental paradigm, analyzing the eye-tracking and behavioral data, and leading the drafting and presentation of the paper.

However, this project only modeled humans' perceptual understanding, whereas real-world decision making extends to a reasoning process, and this in turn guides humans to actively seek additional perceptual information. This led to a new question: *can we model the human perception–reasoning loop so that AI agents can better infer human intent?* Under the guidance of Prof. Shuang Li, I focused on the medical setting, where doctors perceive clinical signals, reason about diagnoses, and then request information from their patients. Doctor's thinking process varies across individuals, and data from a doctor is insufficient to learn a general model. To tackle this, I used a large language model to generate a rule set describing the diagnostic process for a specific disease, and grouped doctors by levels. I then adopted a sparse transformer to model the attention mechanisms of these groups, and implemented a two-stage reinforcement learning procedure to capture the perception $\rightarrow$ reasoning and reasoning $\rightarrow$ perception loop. Our method achieved a 45% diagnostic prediction accuracy on the MIMIC clinical dataset, and this work received the **2025 Undergraduate Research Award** and is in submission to **2026 ICML**.

Physical intelligence around humans

University of Oxford Building on my earlier finding, I next asked: can we equip embodied agents with analogous vision–language mechanisms so that they better understand instructions and produce more reliable actions? Vision–language–action (VLA) models are a natural candidate: they take images and language instructions as input and output actions directly. Under the guidance of Prof. Shimon Whiteson, I set out to investigate how the visual attention mechanisms of VLA models behave and how they affect instruction following. Prior to our work, there was little understanding of how state-of-the-art VLA policies such as Octo [2] and OpenVLA [3] allocate visual attention internally, and we observed that their behavior often degraded on cluttered, multi-object scenes.

To bridge this gap, I developed a visualization pipeline that projects self-attention weights from each layer of the transformer backbone back onto the original camera images. Early layers spread attention broadly; middle layers concentrated on task-relevant objects and the end-effector; but in the final layers, attention became diffuse again. Motivated by this failure mode, I implemented Differential Transformer layers [4], which preserve salient signals by filtering inter-layer attention noise, and integrated them into the final blocks of the VLA backbone. Hence, deeper layers would sharpen vital task features. On the SimplerEnv benchmark [5], the modified model maintained sharp focus on target objects at the deepest layers and achieved a +5.3 percentage point improvement in average success rate. This work has been submitted as a first-author paper to **CVPR 2026**. My contributions included developing the attention visualization pipeline, designing and implementing the Differential Transformer integration, running large-scale simulation experiments, and leading the writing of the paper. Through this work, I began to envision a broader impact—intelligent systems that do not just work in simulation, but also help humans act in the physical world through reliable perception, reasoning, and control. This perspective naturally led me to pursue robot learning under realistic conditions in an industrial research environment.

Microsoft Research Asia I conducted an internship supervised by Dr. Jiang Bian, working on VLA models for real-world manipulation. We evaluated a single model on 10 long-horizon, dexterous tasks with unseen objects, in both simulation and real-robot phases. In the real-robot stage, we initially observed strong jitter in the robot's motions. By inspecting the action sequences, I traced this to large discontinuities between consecutive actions and limited 3D spatial understanding. I proposed two solutions. First, I added a minimum-jerk–inspired regularization term during training to encourage smoother, dynamically feasible trajectories. Second, I incorporated depth information via a spatial distillation loss: a 3D teacher model provided rich spatial features, and the VLA backbone was trained to match these while keeping its original input format. This forced the backbone to internalize 3D structure without destabilizing training. These modifications significantly reduced motion jitter and improved manipulation robustness. Our team ultimately won **second place** at the **2026 IROS Manipulation Challenge**, and I was recognized as one of the **Rising Tech Talents**. This experience deepened my understanding of how to bridge the simulation–reality gap and how to encode basic physical principles directly into large control models.

Future Work Looking ahead, I want to bring my experience in embodied control and attention modeling together with advances in 3D perception to build robots that maintain rich, persistent world models while acting in human environments. I am particularly drawn to Prof. Qianqian Wang's research on reconstructing and tracking dynamic 3D scenes from everyday videos and on continuous 3D perception with persistent state. Her work suggests a compelling path toward agents that not only react to images frame-by-frame, but also maintain a long-horizon understanding of how the world evolves over time.

Concretely, I hope to explore three directions. First, I would like to develop continuous 3D perception models that fuse long-context video, language instructions, and interaction traces into a structured 4D state—one. Building on Prof. Wang's work in 4D reconstruction and tracking, I aim to adapt these representations so that they can be queried and updated by robot policies, enabling agents to reason about occlusions, persistence, and "what-if" futures rather than relying solely on single-view image features. Second, I plan to study active perception: how an embodied agent should move its cameras, end-effector, or even its whole body to resolve uncertainty in this 3D state. Third, I want to connect these 3D world models to foundation policies, investigating how spatially grounded representations can improve generalization to new tasks, objects, and environments.

References

[1] Lichuan Jiang, Jiani Zhong, Muqing Jian, Xuanzhuo Liu, Siqi Cai, and Haizhou Li. The impact of synchronized visual and auditory attention on human perception. In *Social Robotics: 16th International Conference, ICSR + InnoBiz 2024, Shenzhen, China, September 25–28, 2024, Proceedings*, volume 15170 of *Lecture Notes in Computer Science*, pages 41–50. Springer, 2025. URL https://link.springer.com/chapter/10.1007/978-981-96-1151-5_6.

[2] Octo Model Team, Dibya Ghosh, Homer Rich Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Pannag Sanketi, Quan Vuong, Ted Xiao, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. Octo: An open-source generalist robot policy. *arXiv preprint arXiv:2405.12213*, 2024. URL https://arxiv.org/abs/2405.12213.

[3] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024. URL https://arxiv.org/abs/2406.09246.

[4] Tianzhu Ye, Li Dong, Yuqing Xia, Yutao Sun, Yi Zhu, Gao Huang, and Furu Wei. Differential transformer. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025. URL https://proceedings.iclr.cc/paper/2025/file/00b67df24009747e8bbed4c2c6f9c825-Paper-Conference.pdf.

[5] Xuanlin Li, Kyle Hsu, Jiayuan Gu, Karl Pertsch, Oier Mees, Homer Rich Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, and Ted Xiao. Evaluating real-world robot manipulation policies in simulation. *arXiv preprint arXiv:2405.05941*, 2024. URL https://arxiv.org/abs/2405.05941.

* denotes equal contribution.